

Technology Stack

Date	7 Jun 2026
Team ID	
Project Name	OptiCrop
Maximum Marks	4 Marks

Technical Architecture:

The system implementation architecture utilizes machine learning capabilities to evaluate field constraints and deliver crop recommendations.

Table-1: Component Architecture Mapping

S.No	Component	Description	Technology
1	User Interface	Web dashboard interface collecting soil and climate parameters.	HTML, CSS, JavaScript, Streamlit
2	Application Logic-1	Data routing layer and operational parameters logic.	Python, Flask
3	Application Logic-2	Feature scaling, transformation, and array alignment pipeline.	NumPy, Pandas
4	Application Logic-3	Model serialization and un-pickling handling pipeline.	Python Native IO
5	Database	Local tabular structured application logging.	SQLite / MySQL
6	Cloud Database	Remote analytics logging database for historical matrix tracking.	MongoDB Atlas
7	File Storage	Repository for pre-trained serial files (.pkl format matrix).	Local Filesystem
8	External API-1	Environmental current weather parameter verification.	OpenWeatherMap API
9	External API-2	Geographical mapping and territorial verification.	Google Maps Geolocation API
10	Machine Learning Model	Multi-class classification inference algorithm engine.	Random Forest Classifier (Scikit-Learn)
11	Infrastructure (Server / Cloud)	Execution and deployment hosting target.	Local Host / AWS EC2

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology Used
1	Open-Source Frameworks	Elimination of commercial platform dependency fields.	Python Open-Source Data Ecosystem
2	Security Implementations	Input sanitation boundaries checking array limits.	Input Validation Controls
3	Scalable Architecture	Decoupled frontend from background mathematical calculations.	Modular Service Layer Architecture
4	Availability	Stateless hosting components verifying rapid loading cycles.	Stateless local / cloud container profiles
5	Performance	Sub-second localized multi-class distribution predictions.	Lightweight Pickled Array Calculations